When Does Cross-Method Consensus Improve Circuit Discovery? A Systematic Analysis of Failure Modes

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Abstract

Mechanistic interpretability increasingly relies on automated circuit discovery to identify minimal subgraphs that explain model behavior. When multiple discovery methods exist, a natural hypothesis is that their consensus—edges selected by all methods—yields more faithful circuits. We test this hypothesis systematically by applying binary consensus scoring (CSS) across three methods (EAP-IG, Edge Pruning, ACDC) on four tasks and two model scales (GPT-2 Small, 32K edges; Qwen-0.5B, 179K edges). Consensus produces a marginal advantage on only 1 of 5 configurations (IOI on GPT-2, ΔCPR $= 0.08$, $p = 0.09$), which attenuates to $\Delta = 0.02$ ($p = 0.27$) after controlling for circuit size. We identify three distinct failure modes: (1) ACDC's `isinf` binarization collapses consensus to a single method's output, (2) high inter-method agreement (Jaccard > 0.5) renders consensus redundant, and (3) component-level scoring granularity on larger models produces near-zero cross-method overlap. These findings yield practical diagnostic guidelines: practitioners should check pairwise Jaccard similarity and scoring granularity compatibility *before* investing in multi-method consensus pipelines.

1 Introduction

Circuit discovery aims to identify minimal computational subgraphs within neural networks that are sufficient to reproduce a target behavior Conmy et al. (2023); Wang et al. (2023). Several automated methods now exist—gradient-based approaches such as EAP-IG Syed et al. (2023) and Edge Pruning Bhaskar et al. (2024), ablation-based methods such as ACDC Conmy et al. (2023), and hybrid approaches like Iterative Feature Refinement (IFR).

When multiple methods are available for the same problem, ensemble strategies from machine learning suggest that their agreement should be more reliable than any individual method. This intuition is formalized in feature selection stability theory Nogueira and Brown (2016): the intersection of independently selected feature sets should be enriched for genuinely important features. Applied to circuit discovery, this motivates *Cross-method Consensus Scoring* (CSS): binarize each method's output and take their intersection.

We test this hypothesis systematically across four tasks (Indirect Object Identification, Greater-Than Arithmetic, Subject-Verb Agreement, Country-Capital Factual Recall) and two model scales (GPT-2 Small with $\sim$32K edges, Qwen-0.5B with $\sim$179K edges). Our main findings are:

1. CSS produces a marginal faithfulness advantage on only 1 of 5 configurations (IOI on GPT-2), with the advantage largely attributable to circuit size effects (§4.1).
2. We identify three mechanistically distinct failure modes: binarization pathology (§4.3), consensus redundancy under high agreement (§4.4), and scoring granularity mismatch at scale (§4.5).
3. A simple Jaccard-based diagnostic correctly predicts CSS utility in 4 of 5 configurations, though statistical power is insufficient for formal validation (§4.6).

Our contribution is not the consensus framework itself—which follows directly from stability indices Nogueira and Brown (2016); Kuncheva and Whitaker (2003)—but the systematic empirical characterization of *when and why* it fails in the circuit discovery setting, yielding actionable practitioner guidelines.

2 Background and Related Work

Circuit discovery methods. We study three representative methods spanning two paradigms. **EAP-IG** Syed et al. (2023) computes integrated-gradient attributions for each edge in the computational graph. **Edge Pruning** (EP; Bhaskar et al., 2024) optimizes continuous edge masks via gradient descent. Both are gradient-based and score at edge-level granularity. **ACDC** Conmy et al. (2023) iteratively removes edges via activation patching, scoring at node level and propagating scores to edges.

Consensus and ensemble methods. The idea of aggregating multiple discovery methods has been explored in concurrent work. Mondorf et al. (2025) propose cross-method score averaging over continuous attribution values. PIE Chen et al. (2026) uses iterative pruning ensembles within a single paradigm. CIRCUS Parekh et al. (2026) applies within-method threshold diversity rather than cross-method consensus. Our approach differs in using binary consensus (intersection), which enables direct failure mode analysis without conflating score-scale incompatibilities.

Feature selection stability. Binary consensus is an instance of the stability index framework of Nogueira and Brown (2016). The theoretical conditions under which ensemble intersection improves over individual selectors are well-characterized Kuncheva and Whitaker (2003): methods must be (a) individually accurate and (b) sufficiently diverse (low pairwise correlation). Our failure modes map directly onto violations of these conditions.

3 Method: Cross-Method Consensus Scoring

Setup. Given a language model $\mathcal{M}$ with computational graph $G = (V, E)$ where $|E|$ can range from $\sim$32K (GPT-2 Small) to $\sim$179K (Qwen-0.5B), and a set of K circuit discovery methods $\{M_1, \ldots, M_K\}$, each method produces a ranking $r_k : E \to \mathbb{R}$.

Binarization. We convert each continuous ranking to a binary mask. For gradient-based methods (EAP-IG, EP), we select the top-N edges by attribution magnitude. For ACDC, which produces a binary circuit, we use its native output (which we show in §4.3 to be problematic). As a normalization, we also evaluate top-N binarization uniformly across all methods.

Consensus scoring. For each edge $e \in E$, we compute:

$$\text{CSS}(e) = \frac{1}{K} \sum_{k=1}^{K} \mathbf{1}[e \in \text{top-}N(M_k)] \tag{1}$$

Edges with $\text{CSS} = 1.0$ (selected by all K methods) form the consensus circuit. We also examine $\text{CSS} \geq 2/3$ (majority consensus).

Evaluation metrics. We evaluate circuits using **Circuit Performance Recovery** (CPR): the fraction of clean model performance recovered when running only the circuit edges (with remaining edges ablated). We use zero ablation throughout (see §6 for discussion). We also report **Circuit Mean Degradation** ($\text{CMD} = 1 - \text{CPR}$).

Circuit	N	CPR	CMD	Δ
CSS$=1.0$	7	0.109	0.911	—
ACDC top-7	7	0.092	0.928	-0.017
EAP-IG top-7	7	0.078	0.942	-0.031
EP top-7	7	0.065	0.955	-0.044
Random-7	7	$-0.031_{\pm 0.045}$	$1.051_{\pm 0.045}$	-0.140

Table 1: IOI size-controlled comparison ($N = 7$ edges, zero ablation). CSS retains a small advantage over individual methods' top-7 edges, but the margin is substantially reduced from the uncontrolled comparison.

Comparison	Orig. Δ	Size-ctrl Δ	Retained
CSS vs ACDC	0.081	0.017	42%
CSS vs EAP-IG	0.218	0.031	14%
CSS vs EP	0.350	0.044	13%

Table 2: Size confound decomposition for IOI. Controlling for circuit size reduces the CSS advantage by 58–87%, indicating that the majority of the original advantage is attributable to circuit size rather than consensus quality.

4 Experiments

We evaluate CSS with $K = 3$ methods (EAP-IG, EP, ACDC) across four tasks on GPT-2 Small, and additionally test scaling behavior on Qwen-0.5B for IOI.

4.1 IOI: The Marginal Positive Case

Indirect Object Identification (IOI; Wang et al., 2023) is a well-studied task where the model must identify "Mary" in "When John and Mary went to the store, John gave a drink to __."

On GPT-2 Small with top-100 binarization, the CSS$=1.0$ consensus circuit contains only 7 edges. In the original (uncontrolled) comparison, this circuit achieves CPR $= 0.109$, outperforming all single-method top-100 circuits. However, this comparison conflates the consensus effect with circuit size: a 7-edge circuit naturally suffers less ablation damage than a 100-edge circuit.

Size confound analysis. To isolate the consensus effect from circuit size, we compare CSS$=1.0$ (7 edges) against the top-7 edges from each individual method and a random baseline. Table 1 presents the results.

After size control, CSS$=1.0$ retains a residual advantage of ΔCPR $= 0.017$ over the best single method (ACDC top-7). While directionally consistent with the consensus hypothesis, this margin is small and, as we show next, not statistically significant.

Bootstrap confidence intervals. Table 3 presents bootstrap 95% confidence intervals ($B = 10{,}000$ resamples) for CPR differences across all configurations. The size-controlled IOI advantage has a wide CI of $[-0.038, 0.071]$ ($p = 0.27$), failing to exclude zero. Even the original (uncontrolled) IOI advantage is only marginally significant ($p = 0.089$, CI $[-0.012, 0.174]$).

We therefore characterize IOI as a *marginal* positive: CSS shows a directionally correct but statistically weak advantage that is largely attributable to circuit size effects.

4.2 Factual Recall: An Inconclusive Configuration

Country-Capital factual recall on GPT-2 Small tests knowledge retrieval circuits, which are expected to be MLP-heavy. We apply the same $K = 3$ methods with top-100 binarization. Table 4 summarizes the

Task	Comparison	ΔCPR	95% CI	p	Sig.
IOI	CSS vs ACDC (orig.)	0.081	$[-0.01, 0.17]$	.089	No
IOI	CSS vs ACDC (size-ctrl)	0.017	$[-0.04, 0.07]$	.270	No
IOI	CSS vs EAP-IG (orig.)	0.218	$[0.10, 0.34]$	.002	Yes*
Arith.	CSS vs EAP-IG	-0.066	$[-0.14, 0.01]$	.960	No
SVA	CSS vs ACDC	0.013	$[-0.05, 0.08]$	.350	No

Table 3: Bootstrap 95% CIs for CPR differences ($B = 10{,}000$). *Significant but confounded by circuit size (see Table 2).

Circuit	Edges	CPR↑	CMD↓
CSS$= 1.0$	12	0.83	0.41
CSS≥ 0.67	38	0.79	0.48
EAP-IG top-100	100	0.88	0.32
EP top-100	100	0.82	0.43
ACDC top-100	100	0.85	0.37

Table 4: Factual Recall CPR comparison (zero ablation). CSS$= 1.0$ does not outperform the best single method (EAP-IG, CPR $= 0.88$ vs 0.83).

results.

The consensus circuit (CSS $= 1.0$, 12 edges) achieves CPR $= 0.83$, which is *lower* than EAP-IG alone (CPR $= 0.88$, $\Delta = -0.05$). Pairwise Jaccard indices are low overall (EAP-IG×EP: 0.42, EAP-IG×ACDC: 0.18, EP×ACDC: 0.15, three-way: 0.11), with 89% of edges appearing in the off-diagonal region of the CSS distribution.

The failure pattern here differs from other configurations: methods disagree not because of binarization artifacts or scaling issues, but because of *component-type preference*—gradient methods favor attention edges (where gradients through softmax are larger), while ACDC's ablation-based approach better localizes MLP contributions. This creates low Jaccard scores despite each method individually achieving reasonable CPR. The result is inconclusive: CSS neither helps nor catastrophically fails, but cannot recover the complementary information that different methods capture about different component types.

4.3 Failure Mode 1: ACDC Binarization Pathology

ACDC produces circuits via iterative activation patching with an `isinf`-like threshold: edges whose ablation effect exceeds a threshold are retained. This creates a structural asymmetry with gradient-based methods: ACDC's output is already binary and typically much sparser.

Under the standard pipeline, CSS$= 1.0$ edges are necessarily a subset of ACDC's output (since ACDC contributes a binary $\{0, 1\}$ vote). In configurations where ACDC selects fewer edges than the top-N used for gradient methods, the consensus circuit degenerates to a subset of ACDC's circuit. We observed this pattern on IOI (5–6 of 7 consensus edges originate from ACDC), Arithmetic, and SVA.

Switching to uniform top-N binarization across all methods breaks this degeneracy and shifts pairwise Jaccard by $+0.16$ to $+0.25$, confirming that the pathology is an artifact of binarization strategy rather than genuine method agreement.

4.4 Failure Mode 2: Consensus Redundancy

On Greater-Than Arithmetic and Subject-Verb Agreement (SVA), pairwise Jaccard indices are high (Arithmetic: max 0.695; SVA: max 0.653). When methods already agree substantially, consensus provides no new information—the intersection is essentially a single method's output with slightly fewer edges. For Arithmetic, CSS produces $\Delta\text{CPR} = -0.066$ versus the best single method (bootstrap CI

Method	Scoring Level	Eff. Resolution	Unique / 100
EAP-IG	edge	1 edge	100
EP	edge	1 edge	100
IFR	block (layer)	$\sim$200 edges	$\sim$3–5
ACDC	node (head/MLP)	$\sim$500–2K edges	$\sim$8–12

Table 5: Scoring granularity comparison on Qwen-0.5B. Non-gradient methods assign identical scores to all edges within a component, producing block-structured selections with few unique score values.

Pair	Jaccard	Same Gran.?
EAP-IG $\times$ EP	0.54	Yes (edge–edge)
EAP-IG $\times$ IFR	0.00	No (edge–block)
EAP-IG $\times$ ACDC	0.04	No (edge–node)
EP $\times$ IFR	0.00	No (edge–block)
EP $\times$ ACDC	0.04	No (edge–node)
IFR $\times$ ACDC	0.00	No (block–node)

Table 6: Pairwise Jaccard on Qwen-0.5B by scoring granularity. Methods operating at different granularities have near-zero overlap.

$[-0.14, 0.01]$, $p = 0.96$), indicating that the intersection actually *removes* useful edges that not all methods select. SVA shows a near-zero effect ($\Delta = 0.013$, CI $[-0.05, 0.08]$).

4.5 Failure Mode 3: Scoring Granularity Mismatch

On Qwen-0.5B ($\sim$179K edges), we observe a qualitatively different failure: near-zero inter-method overlap regardless of binarization. Root cause analysis reveals that both non-gradient methods degenerate to coarser-than-edge scoring:

IFR assigns identical scores to all edges within a layer block ($\sim$200 edges per block), while ACDC assigns identical scores to all edges from the same attention head or MLP block ($\sim$500–2,000 edges per node). Table 6 shows the resulting pairwise Jaccard values.

This is not an implementation bug but an architectural limitation: IFR's forward intervention is inherently layer-granular, and ACDC's activation patching operates at node granularity. With $K = 3$ methods where the two gradient methods are highly correlated (Jaccard $= 0.54$), the effective number of independent information sources drops from $K = 3$ to $K \approx 1.5$, rendering consensus meaningless.

4.6 Jaccard Diagnostic

Aggregating across all 5 task$\times$model configurations, we observe a negative relationship between maximum pairwise Jaccard and CSS advantage (ΔCPR), as shown in Table 7.

A simple threshold rule emerges: configurations with max Jaccard < 0.35 benefit from CSS (1/1), while those with max Jaccard > 0.50 never do (0/3). The gap zone (0.35–0.50, Factual Recall) yields an incorrect prediction. Spearman $\rho = -0.70$, but with $n = 5$, $p = 0.19$—far from significant.

We frame this as a *motivated conjecture*: the pattern is directionally consistent and mechanistically plausible (low Jaccard implies diverse methods, a prerequisite for useful consensus), but $n = 5$ configurations are insufficient for statistical validation. Practitioners may find the Jaccard check useful as a cheap pre-screening heuristic, but it should not be treated as a validated decision boundary.

4.7 CIRCUS Comparison

As an ablation, we compare CSS (cross-method consensus) against CIRCUS-style within-method threshold diversity: varying the binarization threshold within a single method to generate multiple circuit "vot-

Configuration	Max Jaccard	ΔCPR	CSS Helps?
IOI / GPT-2	0.307	$+0.081$	Yes (marginal)
Factual Recall / GPT-2	0.420	-0.050	No
IOI / Qwen-0.5B	0.540	0.000	No
SVA / GPT-2	0.653	$+0.013$	No
Arithmetic / GPT-2	0.695	-0.066	No

Table 7: Max pairwise Jaccard vs CSS advantage across all configurations. CSS shows a positive effect only in the lowest-Jaccard configuration.

ers." CIRCUS produces negative CPR on all three tested tasks (IOI, Arithmetic, SVA), confirming that cross-method diversity provides signal that within-method threshold variation cannot replicate.

5 Discussion

Failure mode taxonomy. Our three failure modes map onto known conditions from ensemble theory Kuncheva and Whitaker (2003). Mode 1 (binarization pathology) violates the *independence* assumption: one method's binarization dominates the consensus. Mode 2 (high agreement) violates the *diversity* requirement: highly correlated methods provide redundant votes. Mode 3 (granularity mismatch) violates the *commensurability* assumption: methods must operate on the same feature space for intersection to be meaningful.

Practitioner guidelines. Based on our findings, we recommend: (1) normalize binarization across methods (use top-N uniformly rather than mixing native thresholds); (2) compute pairwise Jaccard before investing in consensus—if max Jaccard > 0.5, consensus is unlikely to help; (3) verify that all methods score at the same granularity, especially on models with > 100K edges.

6 Limitations

Size confound. The sole positive configuration (IOI on GPT-2) is substantially confounded by circuit size. In the original comparison, CSS$= 1.0$ (7 edges) is evaluated against single-method top-100 circuits, creating a $14\times$ size disparity. After controlling for size (Table 1), the CSS advantage shrinks from $\Delta = 0.081$ to $\Delta = 0.017$—a 79% reduction. The residual advantage, while directionally correct, is not statistically significant ($p = 0.27$). We cannot rule out that the entire CSS effect on IOI is a size artifact.

Statistical power. Bootstrap confidence intervals are wide throughout: the IOI original-comparison CI spans $[-0.012, 0.174]$, with the lower bound near zero. Our evaluation set (100 prompts $\times$ 7 edges) provides limited statistical power. The Jaccard diagnostic analysis covers only $n = 5$ configurations, insufficient for correlation testing ($p = 0.19$). All quantitative claims should be interpreted as directional evidence, not definitive conclusions.

Task coverage. We evaluate on four tasks (IOI, Arithmetic, SVA, Factual Recall) across two models, totaling 5 complete configurations. This is insufficient to generalize failure mode prevalence or validate the Jaccard threshold. Notably, we have only one configuration with max Jaccard < 0.35, making it impossible to assess CSS's success rate in the low-Jaccard regime.

Ablation method. All evaluations use zero ablation. Zero ablation has known biases for small circuits: ablating $N - k$ edges to zero (where $k \ll N$) can amplify or suppress performance differences relative to mean ablation. We did not complete a systematic comparison of zero vs. mean ablation due to time

constraints. Our results may be sensitive to this choice, particularly the IOI positive finding where the circuit contains only 7 edges.

Method selection. Our $K = 3$ method set includes two gradient-based methods (EAP-IG, EP) that are inherently correlated (Jaccard $= 0.54$ on Qwen-0.5B). This reduces effective diversity. Alternative method sets (e.g., replacing EP with a non-gradient method) might yield different consensus behavior. We chose these methods for their public availability and widespread use, but acknowledge this limits the generality of our negative findings.

7 Conclusion

We presented a systematic evaluation of cross-method binary consensus (CSS) for circuit discovery across four tasks and two model scales. Our main finding is negative: consensus succeeds marginally on only 1 of 5 configurations, with the advantage largely attributable to circuit size effects. We identified three mechanistically distinct failure modes—binarization pathology, consensus redundancy, and scoring granularity mismatch—each with a diagnostic signature. While our task and model coverage is limited ($n = 5$ configurations) and statistical power is low, the failure modes are mechanistically grounded and yield concrete practitioner guidelines.

The broader lesson is that multi-method consensus is not a free lunch in circuit discovery. The conditions under which ensemble approaches improve over individual methods are specific and verifiable. We hope that making these failure modes explicit encourages the community to develop consensus methods that are robust to binarization artifacts, adaptive to method diversity, and aware of scoring granularity—or to invest in improving individual methods rather than ensembling flawed ones.

Future work. Validating the Jaccard diagnostic on a broader set of tasks and models is a priority. A systematic comparison of ablation methods (zero vs. mean) on all configurations would clarify whether the IOI positive finding is robust. Continuous consensus methods Mondorf et al. (2025) that avoid binarization entirely may circumvent Failure Mode 1 and deserve comparative evaluation.

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A Experimental Details

All experiments use the TransformerLens library for model hooks and circuit extraction. For EAP-IG, we use 10-step integrated gradients. For Edge Pruning, we train continuous masks for 500 steps with L_1 regularization ($\lambda = 0.01$). For ACDC, we use the default threshold ($\tau = 0.01$) with iterative node-level patching.

Evaluation uses 100 prompts per task, sampled from the standard evaluation sets. Bootstrap confidence intervals use $B = 10{,}000$ resamples with bias-corrected and accelerated (BCa) percentile intervals.

GPT-2 Small has 12 layers, 12 attention heads, and approximately 32K edges in the computational graph. Qwen-0.5B has 24 layers, 16 attention heads, and approximately 179K edges.